

Vedant K. Naik

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Electrical Engineering student specializing in robotics, embedded systems, and control, with hands-on experience in firmware and system prototyping, seeking opportunities to drive innovation and devise intelligent systems.

EDUCATION

Michigan State University, College of Engineering

Bachelor of Science, Electrical Engineering, GPA: 4.0

East Lansing, MI

August 2022 – May 2026

Activities and Societies: IEEE South East Michigan-Chapter, Tau-Beta-Pi, MSU Solar Racing, PoliMOVE-MSU Autonomous Racing Team

EXPERIENCE

Tesla Motors

Integration Engineering Intern

Palo Alto, CA

August 2025 – December 2025

- Led cross-functional debugging with Motor Controls, Hardware, and Firmware teams to resolve drive-inverter firmware issues and improve system safety and service routine reliability.
- Expanded PIL infrastructure with 2+ motor models and a model-generation pipeline to extend drive-inverter firmware test suites to additional motor variants, improving control evaluation and regression readiness.
- Supported Drive Inverter firmware configurations for a new Drive Unit program, enabling downstream teams to begin validation on schedule and reducing integration delays.

Lucid Motors

Intern, ADAS Sensor Hardware Engineer

Newark, CA

May 2025 – August 2025

- Engineered a camera data collection system for the Lucid Gravity, enabling easy debugging and data collection for all cameras, and providing an I²C interface for rapid prototyping and testing.
- Resolved 5+ production camera issues by analyzing sensor-level signals and raw data, collaborating with ISP vendors, sensor manufacturers, integrators, and internal teams.
- Devised the testing setup for a depth estimation sensing system for the next generation of Lucid vehicles, utilizing a cost-effective approach to repurpose existing camera packaging locations.

Smart Microsystems Lab, Michigan State University

Undergraduate Research Assistant

East Lansing, MI

August 2023 – Present

- Modeled a nonlinear strain sensor with significant temporal dynamics using a Time-Delay Neural Network (TDNN), achieving an $R^2 > 0.98$. Work resulted in a journal publication (DOI: 10.1088/1361-665X/adb2c7).
- Engineered an autonomous robotic blimp with 4-thruster configuration, IMU, WiFi, and LoRa connectivity; developed microcontroller PCB and firmware with micro-ROS architecture to streamline algorithm development between low-level and high-level control systems.

Intrepid Control Systems

Embedded Systems Intern

Troy, MI

May 2024 – August 2024

- Enhanced firmware of Zynq-based products by integrating IMU, DoIP, and GPS support using Xilinx SDK, resolving 7+ firmware driver issues to enrich product functionality.
- Added gPTP support for 10Base-T1S PHY and TC10 support for 100/1000Base-T1 Automotive Ethernet products, optimizing GPS signal reporting and enhancing system performance through the company's resource API.

SKILLS

Embedded Systems & Hardware: Altium Designer (PCB Design), Microcontroller Programming, Soldering, IoT Prototyping, Fusion 360 (CAD), Verilog (HDL), Xilinx SDK, Datasheet Interpretation, Oscilloscope

Programming & Simulation: C, C++, Python, MATLAB, Simulink, git (version control), Linux, Docker

Robotics & AI: ROS 2 (Robot Operating System), OpenCV, TensorFlow (Keras)